# Patrick Thangarajah

Product Design | Prototyping | Testing & Validation | Simulation

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## Summary

Recent MASc graduate in mechanical engineering with practical experience designing, building, and testing hardware systems, including sensor-integrated test setups and control circuitry. Brings simulation, DOE, and data analysis skills to support designs from concept through validation.

## Skills

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| **Design & Prototyping** | SolidWorks, Onshape, Engineering Drawings, 3D Printing, Lathe, Mill, Bandsaw, Drill Press |
| **Project Engineering & Validation** | Testing & validation, DOE, troubleshooting, process improvement, technical problem-solving |
| **Controls & Instrumentation** | Arduino, Sensors, Motors, Data Acquisition, Circuit Design, Altium, LTspice |
| **Simulation & Programming** | FEA, LS-DYNA, LS-PrePost, Python, C/C++ |

## Education

**Master of Applied Science in Mechanical and Mechatronics Engineering** — *Waterloo, ON, Canada*
University of Waterloo — *Sep 2023 - Jul 2026*
- Research in impact mechanics, finite element analysis, and model validation

**Bachelor of Applied Science in Mechanical Engineering, Co-op Program** — *Vancouver, BC, Canada*
University of British Columbia — *Sep 2018 - May 2023*
- Biomechanics and Medical Devices Option

## Experience

**Graduate Research Student - Impact Mechanics and Material Characterization Group** — *Waterloo, ON, Canada*
University of Waterloo — *Sep 2023 - Jul 2026*
- Troubleshot model-to-test discrepancies in positioning, restraint fit, and response timing to improve agreement with physical tests
- Validated simulations against physical tests across two configurations and 10 response channels, achieving a 0.71 average ISO 18571 correlation score
- Ran a 104-configuration LS-DYNA DOE and used JMP to quantify geometry effects on occupant response, identifying a configuration with 16% lower peak chest deflection
- Communicated findings through a master's thesis, conference paper, technical reports, and six sponsor presentations

**R&D Engineer (Part-time)** — *Remote*
Echoscope — *May 2023 - Sep 2023*
- Designed and iterated the bladder testing platform PCB, integrating pressure sensing, Arduino connections, power distribution, and motor and valve interfaces; refined component placement and routing and completed design-rule checks
- Authored five manufacturing and verification procedures covering PCB assembly, wiring, software setup, and functional testing; all five were approved and used by the client

**Mechanical Engineering Intern** — *Airdrie, AB, Canada*
GN Corporation — *Jan 2022 - Apr 2022*
- Created standardized work instructions and SOPs verified against engineering drawings and customer specifications
- Reduced fiberglass cutting waste by 50% through process analysis, and proposed a material handling workflow projected to halve pick-and-place time

**Engineering Intern** — *Victoria, BC, Canada*
Breathe Global — *Jul 2021 - Aug 2021*
- Redesigned the analog control circuit for a negative-pressure breathing device in Altium, using LTspice simulation to select low-cost components
- Expanded the device's achievable breathing range from 70% to 88% of target

## Projects

**Payload Team Member** — *Vancouver, BC, Canada*
UBC Unmanned Aircraft Systems (Student Design Team) — *Sep 2019 - May 2021*
- Designed and prototyped a payload drop system and rover in Onshape and SolidWorks, iterating through 3D printed proof-of-concept builds
- Reduced payload system weight by 50% by repurposing components from previous designs during concept development

**Soft Distensible Bladder Model - Capstone Project** — *Vancouver, BC, Canada*
University of British Columbia — *Sep 2022 - Apr 2023*
- Prototyped bladder phantoms using silicone and agar, comparing pressure-volume response with literature to guide material selection
- Designed and built the control and data-acquisition interface for a pressurized soft-bladder test system, enabling real-time pressure/volume monitoring and automated filling/voiding cycles
- Validated system performance using integrated pressure sensing and controls, demonstrating stable and repeatable operation up to 4.75 kPa internal pressure

## Conference Presentations and Publications

- **Thangarajah, P.** (2026). *Assessment of seat belt anchor position on rear seat occupant kinematics in frontal impact using a validated THOR-5F finite element model*. Master of Applied Science thesis, University of Waterloo, Waterloo, ON, Canada. hdl.handle.net/10012/23684